

Zabbix Monitoring Setup (End-to-End General Guide)

This document provides a **universal step-by-step guide** to set up Zabbix monitoring for any application server (API, web service, or backend service), including agent installation, host configuration, alerting, and optional Grafana integration.

Install Zabbix Agent on Target Server

Purpose

Zabbix agent is installed on target machines to collect system and application metrics.

Install Zabbix repository

```
wget https://repo.zabbix.com/zabbix/7.0/ubuntu/pool/main/z/zabbix-release/  
zabbix-release_latest_7.0+ubuntu22.04_all.deb  
sudo dpkg -i zabbix-release_latest_7.0+ubuntu22.04_all.deb  
sudo apt update
```

Install agent

```
sudo apt install -y zabbix-agent2
```

Enable and start agent

```
sudo systemctl enable zabbix-agent2  
sudo systemctl start zabbix-agent2  
sudo systemctl status zabbix-agent2
```

2 Configure Zabbix Agent

Open config file

```
sudo nano /etc/zabbix/zabbix_agent2.conf
```

Required configuration

Replace values as per your environment:

```
Server=<ZABBIX_SERVER_PRIVATE_IP>  
ServerActive=<ZABBIX_SERVER_PRIVATE_IP>  
Hostname=<UNIQUE_HOSTNAME>
```

Restart agent

```
sudo systemctl restart zabbix-agent2
```

3 Verify Agent Connectivity

From Zabbix server:

```
zabbix_get -s <TARGET_PRIVATE_IP> -k agent.ping
```

Expected output:

```
1
```

4 Create Host in Zabbix UI

Path

Configuration → Hosts → Create host

Host configuration

Field	Value
Host name	Unique hostname
Visible name	Any readable name
Groups	Linux servers / Application servers
Agent interface	<TARGET_PRIVATE_IP>:10050

Link template

- Linux by Zabbix agent (recommended)

5 Create Application Health Check Item (HTTP)

Path

Items → Create item

Configuration

Field	Value
Name	API / Service Health Check
Type	HTTP agent
Key	app.health
URL	http://<TARGET_PRIVATE_IP>:<PORT>/health
Request type	GET
Type of info	Text
Update interval	30s

6 Create Trigger (Failure Detection)

Path

Triggers → Create trigger

Example expressions

Basic failure detection

```
last(/<HOSTNAME>/app.health)<>"ok"
```

Recommended (more reliable)

```
nodata(/<HOSTNAME>/app.health,2m)=1
```

Severity

- High (for production systems)
-

7 Configure Email Notifications (Media Type)

Path

Administration → Media types

SMTP configuration

Field	Value
Type	Email
SMTP server	your-smtp-server
SMTP port	25 / 587
Email sender	zabbix@domain.com
Authentication	As required

8 Add User Media

Path

Users → Select user → Media

Media settings

Field	Value
Type	Email
Send to	user email
Severity	High / Critical

9 Create Action (Trigger → Email)

Path

Configuration → Actions → Trigger actions

Conditions

- Trigger severity = High (or as required)
-

Operations

- Send message to users
 - Send email notification
-

10 Test Monitoring

Start application

```
node app.js
```

Stop application

```
pkill node
```

Expected behavior

State	Result
App running	OK status
App stopped	Trigger problem
Recovery	Trigger resolved

1 1 Grafana Integration (Optional)

Correct API URL

```
http://<ZABBIX_SERVER>/zabbix/api_jsonrpc.php
```

Steps

1. Add Zabbix data source in Grafana
2. Test connection
3. Import dashboards using Grafana.com IDs

Final Architecture

```
Application Server
  ↓
Zabbix Agent (10050)
  ↓
Zabbix Server
  ↓
Trigger Engine
  ↓
Email / Alerts / Grafana Dashboards
```



Summary

This setup provides:

- Infrastructure monitoring
- Application/API health monitoring
- Alerting via email
- Optional visualization via Grafana



Production Tip

Always prefer:

- HTTP status code monitoring OR
- `nodata()` triggers for reliability